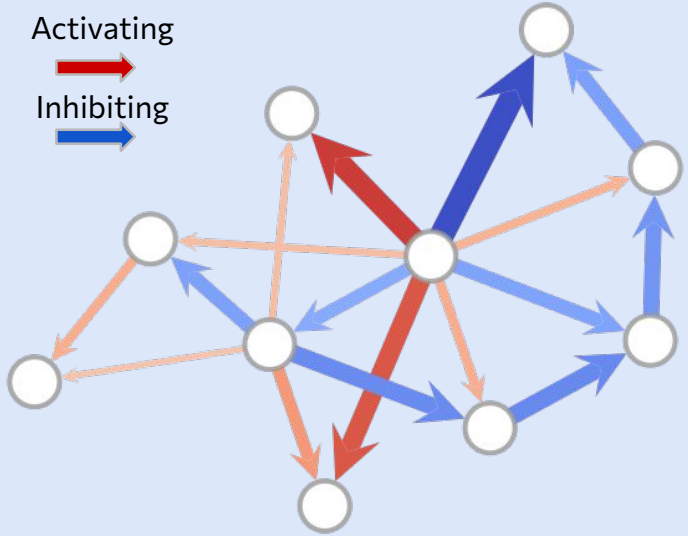
THE MODEL

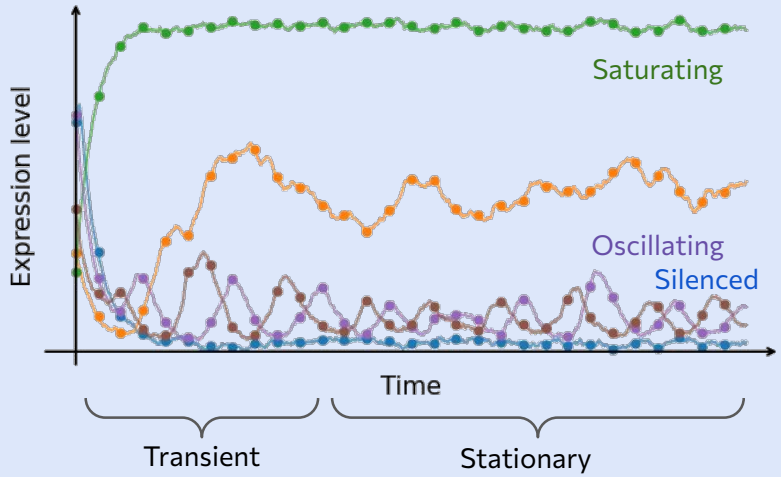


$$dx_i = -\gamma\, x_i + V\sigma\left(\sum_{j=1}^{N} W_{ij}\, x_j\right) + \sqrt{D\, x_i}\, dW_i(t)$$

Deterministic Drift: linear degradation, saturating production

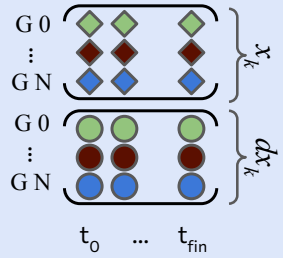
Stochastic term: Multiplicative Noise

THE DATA



Measured are performed at time intervals Δt

Dataset of pairs (x_k, dx_k): from the trajectories we extract the current state and the measured infinitesimal increment



INFERENCE PROCESS

NEQ. Likelihood

$$P(\{\mathbf{x}_k\}_{k=1}^{M}|\Theta)$$

Non-parametric prior

$$P(\mathbf{W} \mid \lambda, \Delta)$$

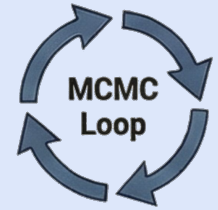
Nelder-Mead minimization

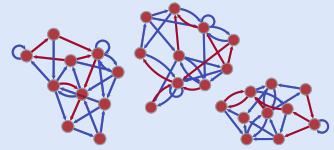


$NN_{Descent}$ Algorithm



MCMC: accept-reject algorithm to explore the posterior distribution





Final ensemble of data-compatible networks